

Assignment 1, Harmonic Analysis

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Problem 1

(a)

Proof.

$$\begin{aligned} M'_c(f)(x) &= \sup_{r>0} \frac{1}{r^d} \int_{x+(-r/2, r/2)^d} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{r^d} \int_{B(x, r\sqrt{d}/2)} |f(y)| dy && (x + (-r/2, r/2)^d \subseteq B(x, r\sqrt{d}/2)) \\ &= \sup_{\rho>0} \frac{d^{d/2}}{2^d \rho^d} \int_{B(x, \rho)} |f(y)| dy && (\rho = r\sqrt{d}/2) \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \sup_{\rho>0} \frac{1}{v_d \rho^d} \int_{B(x, \rho)} |f(y)| dy \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \cdot M_c(f)(x). \quad \square \end{aligned}$$

(b)

Proof.

$$\begin{aligned} M_c(f)(x) &= \sup_{r>0} \frac{1}{v_d r^d} \int_{B(x, r)} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{v_d r^d} \int_{x+(-r, r)^d} |f(y)| dy && (B(x, r) \subseteq x + (-r, r)^d) \\ &= \sup_{s>0} \frac{2^d}{v_d s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy && (s = 2r) \\ &= 2^d \cdot v_d^{-1} \sup_{s>0} \frac{1}{s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy \\ &= 2^d \cdot v_d^{-1} \cdot M'_c(f)(x). \quad \square \end{aligned}$$

Problem 2

(a)

Proof. Since clearly $|f(x) - q| \in L^1_{\text{loc}}(\mathbb{R}^d)$, by Lebesgue Differentiation Theorem

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy = |f(x) - q| \quad \text{for a.e. } x \in \mathbb{R}^d. \quad (1)$$

Let A_q be the set where it fails, then $|A_q| = 0$ and the limit holds for all $x \in \mathbb{R}^d \setminus A_q$. $\square$

(b)

Proof. Let $A := \bigcup_{q \in \mathbb{Q}} A_q$, where each A_q is as in previous part. Since $\mathbb{Q}$ is countable, $|A| = 0$, and for every $x \in \mathbb{R}^d \setminus A$, (1) holds simultaneously for all $q \in \mathbb{Q}$.

Fix $x \in \mathbb{R}^d \setminus A$ and $\varepsilon > 0$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, choose $q \in \mathbb{Q}$ such that $|f(x) - q| < \varepsilon/2$. Because of (1), there exists $r_0 > 0$ such that for all $r \in (0, r_0)$,

$$\frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy < \frac{\varepsilon}{2}.$$

Then for all $r \in (0, r_0)$,

$$\begin{aligned} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| dy &\leq \frac{1}{|B(x, r)|} \int_{B(x, r)} (|f(y) - q| + |f(x) - q|) dy \\ &= \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy + |f(x) - q| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary,

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| dy = 0 \quad \forall x \in \mathbb{R}^d \setminus A. \quad (2) \quad \square$$

(c)

Proof. Let A be the set from previous part and let $r_n := \sqrt{d} \cdot 2^{-n}$, so $Q_n(x) \subseteq B(x, r_n)$, with $|Q_n| = 2^{-nd}$ and $|B(x, r_n)| = v_d r_n^d = v_d d^{d/2} \cdot 2^{-nd}$.

Fix $x \in \mathbb{R}^d \setminus A$ and $\varepsilon > 0$; by (2), $\exists N \in \mathbb{N}$ such that $\forall n \geq N$,

$$\frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy < \frac{\varepsilon}{v_d d^{d/2}}.$$

Then for all $n \geq N$,

$$\begin{aligned}
\left| \frac{1}{|Q_n|} \int_{Q_n} f(y) dy - f(x) \right| &\leq \frac{1}{|Q_n|} \int_{Q_n} |f(y) - f(x)| dy \\
&\leq \frac{1}{|Q_n|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= \frac{|B(x, r_n)|}{|Q_n|} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= \frac{v_d r_n^d}{2^{-nd}} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= \frac{v_d \cdot d^{d/2} \cdot 2^{-nd}}{2^{-nd}} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= v_d d^{d/2} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&< v_d d^{d/2} \cdot \frac{\varepsilon}{v_d d^{d/2}} = \varepsilon.
\end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $\left| \frac{1}{|Q_n|} \int_{Q_n} f(y) dy - f(x) \right| \rightarrow 0$ as $n \rightarrow \infty$, which is equivalent to $\frac{1}{|Q_n(x)|} \int_{Q_n(x)} f(y) dy \rightarrow f(x)$, $\forall x \in \mathbb{R}^d \setminus A$. $\square$

Problem 3

(a)

Proof. For $\delta > 0$, let $f_\delta(x_1, x_2) := \delta^{-2} \chi_{(-\delta, \delta)}(x_1) \chi_{(-\delta, \delta)}(x_2)$. Then $\|f_\delta\|_{L^1} = \delta^{-2} \cdot |(-\delta, \delta)^2| = \delta^{-2} \cdot (2\delta)^2 = 4$. For $x = (x_1, x_2)$ with $x_1, x_2 > \delta$, define the rectangle $R_x^\delta := [-\delta, x_1] \times [-\delta, x_2] \in \mathcal{R}$, with $|R_x^\delta| = (x_1 + \delta)(x_2 + \delta)$.

Suppose for contradiction there exists $C > 0$ such that $|\{M_S(f) > \alpha\}| \leq \frac{C}{\alpha} \|f\|_{L^1}$ for all $f \in L^1(\mathbb{R}^2)$ and $\alpha > 0$. Fix $\alpha > 0$ and choose $\delta_0 < e^{-(C+4)/8} \alpha^{-1/2}$. Then:

$$\begin{aligned}
|\{M_S(f_{\delta_0}) > \alpha\}| &= 4 |\{x_1, x_2 > 0 : M_S(f_{\delta_0})(x_1, x_2) > \alpha\}| && \text{(symmetry, 4 quadrants)} \\
&\geq 4 |\{x_1, x_2 > \delta_0 : M_S(f_{\delta_0})(x_1, x_2) > \alpha\}| \\
&\geq 4 |\{x_1, x_2 > \delta_0 : \frac{1}{|R_x^{\delta_0}|} \int_{R_x^{\delta_0}} f_{\delta_0} > \alpha\}| \\
&= 4 \left| \left\{ x_1, x_2 > \delta_0 : \frac{4}{(x_1 + \delta_0)(x_2 + \delta_0)} > \alpha \right\} \right| \\
&= 4 |\{u, v > 2\delta_0 : uv < \frac{4}{\alpha}\}| && (u = x_1 + \delta_0, v = x_2 + \delta_0) \\
&= 4 \int_{2\delta_0}^{\infty} \left| \{v > 2\delta_0 : v < \frac{4}{\alpha u}\} \right| du \\
&= 4 \int_{2\delta_0}^{2/(\alpha\delta_0)} \left(\frac{4}{\alpha u} - 2\delta_0 \right) du \\
&= \frac{16}{\alpha} \ln \frac{1}{\alpha\delta_0^2} - \frac{16}{\alpha} + 16\delta_0^2 > \frac{4C}{\alpha},
\end{aligned}$$

contradicting $|\{M_S(f_{\delta_0}) > \alpha\}| \leq \frac{C}{\alpha} \|f_{\delta_0}\|_{L^1} = \frac{4C}{\alpha}$. Hence M_S is not of weak-type $(1, 1)$. $\square$

(b)

Proof. For $g \in L^1_{\text{loc}}(\mathbb{R})$, let

$$M(g)(x) := \sup_{\substack{\delta > 0 \\ |x-y| < \delta}} \frac{1}{|B(y, \delta)|} \int_{B(y, \delta)} |g(t)| dt.$$

Since M is weak-type $(1, 1)$ and bounded on $L^\infty(\mathbb{R})$, Marcinkiewicz gives $\|M(g)\|_{L^p(\mathbb{R})} \leq C_p \|g\|_{L^p(\mathbb{R})}$ for $1 < p < \infty$; at $p = \infty$ this holds trivially, with constant 1, directly from the definition. So $\|M(g)\|_{L^p(\mathbb{R})} \leq C_p \|g\|_{L^p(\mathbb{R})}$ for all $1 < p \leq \infty$.

For $t \in \mathbb{R}$, write $f_t(s) := f(t, s)$. By Fubini, M applied in either coordinate direction is bounded on $L^p(\mathbb{R}^2)$ with the same constant C_p , for $1 < p \leq \infty$.

$$\begin{aligned} \|M_S(f)\|_{L^p} &= \left\| \sup_{I_1 \times I_2 \ni x} \frac{1}{|I_1 \times I_2|} \int_{I_1 \times I_2} |f| \right\|_{L^p} \\ &= \left\| \sup_{I_1 \times I_2 \ni x} \frac{1}{|I_1|} \int_{I_1} \frac{1}{|I_2|} \int_{I_2} |f_t(s)| ds dt \right\|_{L^p} \\ &\leq \left\| \sup_{I_1 \ni x_1} \frac{1}{|I_1|} \int_{I_1} M(f_t)(x_2) dt \right\|_{L^p} \\ &= \|M(M(f_t)(x_2))(x_1)\|_{L^p} \\ &\leq C_p \|M(f_t)(x_2)\|_{L^p} \\ &\leq C_p^2 \|f\|_{L^p}. \end{aligned}$$

□